Strategic Framing of Artificial Intelligence: A Cross-Context Corpus Study

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Abstract

AI industry actors communicate simultaneously with investors, regulators, and the general public — yet whether they strategically adapt their framing of AI across these audiences remains empirically untested. We present a cross-context framing analysis of 5,946 documents from 16 major actors spanning commercial, policy, and public communication contexts. We define five binary framing dimensions and label 63,546 sentences using Claude Haiku, achieving inter-annotator agreement $\kappa = 0.86$. OLS regressions on document-level framing proportions confirm that context significantly predicts all three framing DVs ($p < 0.001$, H1 confirmed): policy documents carry 13.6 percentage points more regulation framing than commercial documents. Satya Nadella's policy communications contain 27.1 percentage points more innovation framing than his commercial baseline — the largest individual strategic adaptation signal in the corpus (H2 partially confirmed). These results demonstrate that AI discourse is a strategic communication tool shaped by audience, institutional role, and actor identity.

1 Introduction

When Sam Altman testified before the US Senate Judiciary Committee in May 2023, he emphasized existential risk and the need for regulation. Speaking at a startup conference weeks later, he emphasized transformative economic opportunity. The same actor, the same technology, two radically different framings — chosen for different audiences.

Framing is the selective emphasis of certain aspects of a topic to promote a specific interpretation (Entman, 1993; Goffman, 1974). We test whether AI actors adapt their framing strategically across three communication contexts: commercial (investors and customers), policy (regulators and legislators), and public (general audiences).

Research question. To what extent do actors adapt their framing of AI across commercial, policy, and public contexts, controlling for time, platform, and actor type?

H1 Context significantly predicts framing ($\beta_{\text{context}} \neq 0$).

H2 Commercial contexts increase innovation framing; policy contexts increase risk and regulation framing.

H3 Individual actors show greater cross-context framing variance than institutions.

We make three contributions: (1) a 5,946-document cross-context AI framing corpus spanning 16 actors and three communication contexts; (2) a validated five-frame annotation scheme with $\kappa = 0.86$; and (3) OLS regression evidence that context, actor positioning, and platform independently and significantly predict framing proportion.

2 Related Work

Entman (1993) defines framing as the selection and salience of particular aspects of reality to promote a specific interpretation. Goffman (1974) grounds this in communicative interaction: frames are organizational structures actors deploy to define situations for specific audiences. Chong and Druckman (2007) extend this to strategic political communication, showing that actors choose frames to maximize persuasive impact on targeted audiences — the theoretical foundation for H2 and H3.

Card et al. (2015) demonstrate that frame categories can be annotated at sentence level with reliable inter-annotator agreement across policy issues; their corpus is the closest methodological precedent for our design. We extend their approach to a cross-actor, multi-context setting and focus specifically on strategic adaptation rather than static framing prevalence.

Fast and Horvitz (2017) analyze 30 years of AI coverage in the *New York Times*, finding systematic shifts from optimistic to cautious framing as research milestones accumulate. We complement this longitudinal view with a cross-actor design that isolates deliberate strategic variation within a single time window. Gilardi et al. (2023) show that LLMs match or exceed crowd-worker annotation quality; we extend this finding to a technical domain with expert-level frame definitions, validating against a human gold set and three alternative models.

3 Data

3.1 Corpus Design

We select 16 actors spanning the full range of AI discourse: six CEOs (Sam Altman, Dario Amodei, Jensen Huang, Satya Nadella, Mark Zuckerberg, Demis Hassabis), six companies (OpenAI, Anthropic, Google DeepMind, Meta AI, Microsoft, Nvidia), and four government policymakers (EU Commission, US Congress, UK DSIT, White House OSTP). Each individual is paired with their own company to enable direct individual-vs-institution comparison for H3. Elon Musk was excluded after only 21 accessible documents were found (xAI blocked by Cloudflare; no policy corpus); Satya Nadella replaces him with full three-context coverage.

We assign each actor a **positioning** value — *capability* (Altman, Nadella, Zuckerberg, OpenAI, Meta AI, Microsoft), *safety* (Amodei, Hassabis, Anthropic, Google DeepMind), *infrastructure* (Huang, Nvidia), or *regulator* (all four policymakers) — to test whether institutional identity predicts framing beyond context in Model 3.

3.2 Collection and Corpus Statistics

We build a Python scraping pipeline with `requests` and `BeautifulSoup` across 30+ source types: company blogs, arXiv (author-name queries), government archives, podcast transcripts, and the Wayback Machine CDX API (for sites blocked or deleted from the live web). We manually ingest 21 PDFs via `pdfplumber`. After URL-based deduplication, we apply five sentence-level quality filters (non-English, length $<$8 words, JSON/HTML fragments, no AI-keyword relevance, off-topic patterns) that reduce 208,320 raw sentences to 63,546 clean sentences (30.5% retained).

Table 1 shows the final corpus of 5,946 documents. The public context reaches only 4.0% (target $\geq$15%): YouTube captions are disabled for all 14 targeted CEO videos, podcast platforms use JavaScript rendering, and conference transcripts are paywalled — a structural ceiling, not a sampling choice. Company documents represent 52.3% because $\sim$1,300 arXiv papers are classified as commercial (official company-authored research output), inflating company share.

Table 1: Final corpus distribution (N = 5,946 documents).

Context	Docs	Share
Commercial	3,661	61.6%
Policy	2,048	34.4%
Public	237	4.0%
Total	**5,946**	**100%**
Actor type		
Company	3,111	52.3%
Policymaker	1,704	28.7%
Individual	1,122	18.9%

Table 2: Inter-annotator agreement by annotation round.

Round	κ	Result	Root cause
1	0.36	FAIL	None/frame boundary underspecified
2	0.37	FAIL	Corpus noise
3	0.86	PASS	Clean sentence pool

4 Annotation

4.1 Frame Definitions

We define five binary framing dimensions derived from the computational framing literature (Card et al., 2015) plus a NONE label. Each sentence receives one or more labels, or NONE if no frame is explicitly stated in the text (inferred frames are coded NONE).

Innovation/Progress: AI as transformative, with an explicit claim about societal benefit or scientific advance. *Economic Benefit*: explicit claim about jobs, revenue, productivity, or national competitiveness. *Risk/Harm*: concrete near-term harms — bias, job loss, misuse, or surveillance. *Regulation/Governance*: explicit institutional mechanism — law, framework, oversight body. *Existential/AGI*: long-term civilizational risk, AGI, or superintelligence.

Full definitions and edge-case rules are in `docs/annotation_guidelines.md`.

4.2 Inter-Annotator Agreement

Two annotators independently label a stratified 100-sentence overlap set (balanced across contexts and actors), measuring Cohen's κ (Landis and Koch, 1977) with a $\kappa \geq 0.70$ gate before corpus-scale labeling.

Rounds 1 and 2 fail despite guideline revision after Round 1. Diagnostic analysis reveals that 24 of 28 Round 2 disagreements trace to identical problematic sentences: React/Next.js JSON fragments, FEMA boilerplate, three-word header fragments, and economic statistics with no AI content. We apply five quality filters (§3.2) to remove 144,774 sentences, redraw the overlap set from the clean pool, and run Round 3, achieving $\kappa = 0.86$ across all six label categories (Table 2). The single-round jump confirms that early failures reflect corpus noise, not genuine annotation ambiguity.

4.3 LLM Labeling and Validation

We label all 63,546 sentences using `claude-haiku-4-5-20251001` via the Anthropic Messages API, using a system prompt derived directly from the annotation guidelines (batch size 15, 0.3-second inter-call delay). We validate against the 100-sentence human gold set: macro precision = 0.847, recall = 0.534, F1 = 0.633. High precision indicates that when Haiku assigns a frame, it is almost always correct; low recall indicates it defaults to NONE for $\sim$47% of true positives. Framing scores are therefore systematic underestimates — positive findings are *conservative*, not inflated. Three alternative models confirm the same conservative bias pattern: GPT-4o-mini F1 = 0.472, Llama 3.3-70B F1 = 0.415, Gemini 2.5-Flash F1 = 0.214. Haiku achieves the highest F1 of any tested model.

4.4 Dependent Variable Selection

We exclude two frames as regression DVs. **Economic Benefit** has LLM recall = 0.00 in the policy context, making any coefficient an artifact of annotation failure rather than real framing. **Existential/AGI** reaches only 1.1% sentence frequency with near-zero document-level variance. The three retained DVs — *innovation_score*, *risk_score*, *regulation_score* — have frequencies of 18.7%, 8.7%, and 11.5% and non-zero LLM recall in every context.

5 Results

For each document d and frame F: $\text{score}_{F,d} = (\text{sentences labeled } F \text{ in } d)/(\text{total sentences in } d)$. This proportion normalizes framing density by document length. Documents with fewer than 5 sentences are excluded (3,411 removed), yielding $N = 2{,}535$ documents for regression. Reference categories: context = commercial, positioning = capability, platform = blog, actor_type = company.

5.1 H1: Context Predicts Framing (Model 1)

Model 1: $\text{score}_d = \beta_0 + \beta_1 C(\text{context}_d) + \beta_2 \text{post\_chatgpt}_d + \varepsilon_d$. Post-ChatGPT is a binary equal to 1 for documents dated on or after 2022-11-30, controlling for corpus-wide discursive intensification after the public launch of ChatGPT.

H1 is confirmed. Table 3 shows that context is a significant predictor of all three DVs. Policy documents carry 13.6 pp more regulation framing than commercial ($\beta = +0.136$, $p < 0.001$), explaining 12.1% of regulation variance. Risk framing peaks in policy (+5.5 pp) and drops in public settings (-4.3 pp). Innovation is significantly lower in policy than commercial (-2.3 pp, $p = 0.016$), confirming commercial as the innovation register by inversion. Post-ChatGPT intensification is significant for innovation (+4.8 pp) and regulation (+2.3 pp), reflecting a corpus-wide discursive shift after November 2022. The low R^2 for innovation (0.014) reveals substantial actor-level heterogeneity that requires Model 2.

Table 3: Model 1 OLS results ($N = 2{,}535$). Coefficients are pp changes relative to commercial baseline. SE in parentheses. ***<0.001, **<0.01, *<0.05.

Predictor	**Innov.**	**Risk**	**Reg.**
Policy	-0.023^{*}	$+0.055^{***}$	$+0.136^{***}$
	(0.009)	(0.006)	(0.008)
Public	$+0.036$	-0.043^{***}	-0.017
	(0.019)	(0.012)	(0.016)
Post-ChatGPT	$+0.048^{***}$	$+0.002$	$+0.023^{**}$
	(0.009)	(0.006)	(0.007)
R^2	0.014	0.041	0.121

Table 4: Model 2: significant actor×policy interactions ($N = 636$). Reference actor: Microsoft.

Actor	**DV**	$\hat{\beta}$	**SE**	p
Nadella	innovation	$+0.271$	0.044	$<0.001^{***}$
OpenAI	innovation	$+0.101$	0.037	0.007^{**}
OpenAI	regulation	-0.097	0.023	$<0.001^{***}$
OpenAI	risk	-0.056	0.028	0.046^{*}

5.2 H2: Strategic Adaptation (Model 2)

Model 2 adds actor fixed effects and actor×context interactions. A significant positive interaction means the actor frames more intensely in that context than actor-level and context-level effects alone predict — the operational definition of strategic adaptation.

Model 2 requires $\geq$50 documents per actor×context cell and presence in $\geq$2 contexts. Three actors qualify: Microsoft (comm $n = 84$, pol $n = 152$), OpenAI (comm $n = 194$, pol $n = 81$), and Satya Nadella (comm $n = 59$, pol $n = 66$), yielding $N = 636$ documents. Public context is excluded — sparse coverage produces near-degenerate estimates.

H2 is partially confirmed. Table 4 shows four significant interactions. Satya Nadella's policy documents contain 27.1 pp more innovation framing than his com-

Table 5: Cross-context variance (σ) for qualifying actors. Higher σ = greater framing adaptation across contexts.

Actor	Type	Ctx	Inn. σ	Risk σ	Reg. σ
Satya Nadella	ind	3	**0.129**	0.050	**0.064**
Microsoft	co	3	0.031	0.049	0.049
UK DSIT	pm	2	0.088	0.036	0.012
OpenAI	co	2	0.037	0.007	0.011
Google DeepMind	co	2	0.011	0.032	0.016

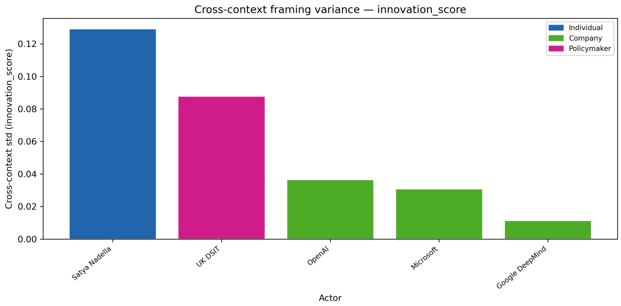

Figure 1: Cross-context innovation variance for qualifying actors. Nadella (individual) shows substantially higher variance than all qualifying companies. Variance figures for risk and regulation are in `outputs/figures/`.

mercial baseline plus the average policy shift predicts ($\beta = +0.271$, $p < 0.001$) — the strongest strategic adaptation signal in the corpus. When addressing regulators, Nadella foregrounds AI as an innovation driver rather than a governance challenge. OpenAI increases innovation framing in policy (+10.1 pp) while simultaneously suppressing regulation (-9.7 pp) and risk (-5.6 pp) language, suggesting active resistance to regulatory framing when communicating with policymakers.

The H2 innovation finding is conservative: LLM recall for Innovation/Progress is 28 pp lower in commercial contexts (0.28) than in policy (0.56), meaning the measured commercial > policy gap understates the true effect. Economic Benefit is untestable given LLM recall = 0.00 in policy.

5.3 H3: Individual vs. Institution Variance

For each qualifying actor ($\geq$50 documents in $\geq$2 contexts) we compute the standard deviation of context-level mean framing scores. Five of 16 actors qualify.

H3 is directionally supported but not formally testable. Satya Nadella — the sole qualifying individual — exhibits the highest cross-context variance on innovation ($\sigma = 0.129$) and regulation ($\sigma = 0.064$), 4.2× and 1.3× higher than paired company Microsoft (Table 5; Figure 1). A formal t-test requires $\geq$2 qualifying individuals; only 1 qualifies due to the public-context coverage gap. We report Nadella's result as illustrative evidence consistent with H3, with corpus limitations explicitly disclosed. The M2 interaction ($\beta = +0.271^{***}$) and the variance analysis ($\sigma = 0.129$) are complementary views of the same pattern.

5.4 Model 3: Structural Controls

Model 3 adds actor type, positioning, platform, and time (N = 2,535). The cluster {policymaker, policy context, regulator positioning, testimony platform} is near-perfectly collinear; we report only interpretable terms. *Safety* positioning predicts less innovation framing ($\beta = -0.042$, $p < 0.001$), more risk ($\beta = +0.038$, $p < 0.001$), and more regulation ($\beta = +0.050$, $p < 0.001$) — confirming that actors' public positioning maps onto their linguistic output across all contexts. Speeches carry 14.5 pp more innovation framing than blogs ($\beta = +0.145$, $p = 0.006$); research papers suppress both innovation ($\beta = -0.084$, $p < 0.001$) and regulation framing ($\beta = -0.031$, $p = 0.007$). Regulation framing achieves the highest explained variance across all models ($R^2 = 0.221$ in M3), driven by positioning and platform rather than context alone.

6 Discussion

Satya Nadella adapts more than any other qualifying actor: a 27.1 pp innovation spike in policy documents signals deliberate rhetorical strategy — framing AI as an economic opportunity rather than a governance challenge when addressing Congress. OpenAI shows the inverse: active suppression of regulation and risk language in policy contexts while amplifying innovation framing. Together, these patterns suggest that commercial actors treat policy audiences as arenas for shaping regulatory narrative toward favorable outcomes, not as targets for safety reassurance.

Safety positioning predicts framing content independently of context: Anthropic, Amodei, Hassabis, and Google DeepMind consistently use more risk and regulation language regardless of setting. Institutional identity constrains — but does not eliminate — context-driven adaptation. Actors frame strategically, but within the space their brand positioning defines: a safety-positioned actor does not suppress risk framing even in commercial contexts.

These findings reframe AI discourse as a strategic rather than descriptive practice. Public deliberation about AI policy is not driven by technological facts alone but by deliberate communication strategies tuned to audience expectations and institutional incentives. The public-context gap prevents formal H3 testing;

richer public-context data would resolve whether individual CEOs adapt more freely than their companies.

7 Conclusion

We study whether major AI actors strategically adapt their framing across commercial, policy, and public communication contexts. We construct a 5,946-document corpus, annotate 63,546 sentences with $\kappa = 0.86$, and estimate OLS regression models on document-level framing proportions. H1 is confirmed: context significantly predicts all three framing dimensions ($p < 0.001$). H2 is partially confirmed: policy increases risk (+5.5 pp) and regulation (+13.6 pp) framing; Satya Nadella and OpenAI exhibit strong strategic innovation framing in policy contexts. H3 is directionally supported but not formally testable given that only one individual qualifies for variance analysis. Future work should expand public-context coverage — targeted collection of conference keynotes and formally transcribed interviews — to enable direct individual-vs-institution hypothesis testing.

Limitations

Public context underrepresentation. Public documents constitute 4.0% of the corpus (target: $\geq 15\%$). YouTube captions are disabled for all 14 targeted CEO videos; podcast platforms use JavaScript rendering; conference transcripts are paywalled. This structural ceiling collapses the three-context design for most individual actors and prevents formal H3 testing.

arXiv inflation. Company documents represent 52.3% because $\sim$1,300 arXiv papers are classified as commercial (correct: company-authored research serves the same audience function as product blogs) but inflates company share relative to the other actor types.

Actor-context sparsity. Twenty-four actor$\times$context pairs fall below the 50-document minimum. Model 2 is restricted to three actors (Microsoft, OpenAI, Nadella) and cannot generalize to the full corpus. Per-actor counts appear in Appendix A.

English-only corpus. All documents are in English. Multilingual framing variation — which may differ systematically across regulatory jurisdictions — is outside the scope of this study.

Ethical Considerations

All documents are publicly available: corporate blog posts, government publications, official testimonies, and publicly accessible podcast transcripts. No personal data are collected. Wayback Machine sources consist exclusively of previously public pages recovered for academic reproducibility. The LLM is used for annotation only, not to generate any corpus content. All annotation guidelines, prompts, and code are released with the data.

Data Availability

All code, annotation guidelines, regression outputs, and labeled data are available at `https://github.com/juliavikr/ai-framing-project`. All sources were publicly accessible at collection time. See Appendix A for the full per-actor source list.

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A Per-Actor Document Counts

Table 6 reports per-actor counts in the regression dataset ($N = 2{,}535$, after the $\geq$5-sentence filter). Actors with fewer than 50 documents in a given context are excluded from Model 2 interaction analysis. Positioning: C = capability, S = safety, I = infrastructure, R = regulator.

Table 6: Per-actor document counts in the regression dataset (N = 2,535).

Actor	Type	Pos.	Com.	Pol.	Pub.	Tot.
Dario Amodei	ind	S	310	2	3	315
Satya Nadella	ind	C	59	66	36	161
Demis Hassabis	ind	S	80	3	8	91
Mark Zuckerberg	ind	C	118	1	3	122
Jensen Huang	ind	I	148	0	7	155
Sam Altman	ind	C	13	1	3	17
Anthropic	co	S	265	2	7	274
OpenAI	co	C	194	81	0	275
Google DeepMind	co	C	183	0	17	200
Meta AI	co	C	170	0	0	170
Microsoft	co	C	84	152	28	264
Nvidia	co	I	78	0	3	81
UK DSIT	pm	R	0	236	16	252
White House OSTP	pm	R	0	63	0	63
EU Commission	pm	R	0	90	2	92
US Congress	pm	R	0	3	0	3
Total			**1,702**	**700**	**133**	**2,535**